

CONTROLLED TRIAL OF SOYA-BEAN OIL IN MYOCARDIAL INFARCTION

REPORT OF A RESEARCH COMMITTEE TO THE MEDICAL RESEARCH
COUNCIL*

Summary A trial is reported of the value of a serum-cholesterol-lowering diet in the prevention of relapse in men aged under sixty recently recovered from a first myocardial infarction and discharged from four district hospitals in the north and west of London. 199 men randomly allocated to the experimental group were given a diet low in saturated fats and containing 85 g. of soya-bean oil daily; the 194 control patients took their ordinary diet. Both groups weighed their food regularly and a high degree of cooperation was achieved. The shortest individual period in the trial was two years and the longest six and three-quarter years. About half of the men were in the trial for four years or more. The test diet lowered the serum-cholesterol from a mean initial figure of 272 to 213 mg. per 100 ml. at six months (22% fall); the level in the controls fell from 273 to 259 mg. per 100 ml. (6% fall). Suspected relapses were assessed at regular intervals by a review committee unaware of the patient's diet group. 62 men on the test diet suffered at least one relapse during the period of observation compared with 74 of the controls. 40 of the first relapses in the test group were major (i.e., definite reinfarctions or deaths from coronary heart-disease) compared with 39 major first relapses in the controls. The total number of men who had a major relapse at any time in the trial was 45 in the test group and 51 in the controls; of these major relapses 25 in each group were fatal. None of the differences found is significant. Relapses were not related to initial cholesterol level, to change in cholesterol level during the trial, nor, in any consistent way, to observance of the dietary regimen. The results are compared with those from a similar trial in Oslo. There is no evidence from

the London trial that the relapse-rate in myocardial infarction is materially affected by the unsaturated fat content of the diet used.

Introduction

WE report here a controlled dietary study carried out for two to seven years on 393 male patients who had recovered from their first myocardial infarction, and who were discharged from four large district general hospitals in the north and west of London (Central Middlesex, West Middlesex, Edgware General, and Barnet General). The trial lasted from 1960 to 1967, and patients were entered up to March, 1965.

The objective of the trial was to determine whether a diet in which saturated fats were replaced by polyunsaturated fats would reduce the incidence of myocardial reinfarction in men who had survived a first attack.

Selection, Management, and Methods

SELECTION

Criteria for Inclusion

All men, discharged from hospital after admission for a first infarction, and aged under sixty at the time of the infarct were considered for entry to the trial. First infarction was diagnosed on a first history of pain with Q-wave evidence of infarction on the electrocardiogram (E.C.G.), or a history of at least one hour's rest pain plus S-T changes on E.C.G. plus previously agreed ancillary evidence (fever, leucocytosis, raised erythrocyte-sedimentation rate, or raised enzyme levels).

Criteria for Exclusion

73 patients who had fulfilled the above criteria were excluded for the following previously agreed reasons: gross obesity (2); diabetes diagnosed on routine urine testing (8); syphilis diagnosed by positive Wassermann reaction (1); hypertension where the diastolic pressure was consistently over 110 mm. Hg or if potent hypotensive drugs were required (6); patients with cardiac lesions which significantly prejudice the prognosis (e.g., complete heart block, gross cardiomegaly cardiac aneurysm, or ruptured ventricular septum) (7); previous significant modification of dietary fat intake (2); inability of the patient to understand or comply with the requirements of the diet regimen (14); unsuitability for prolonged anticoagulant therapy (e.g., patients on steroid therapy or with peptic ulceration or hæmorrhoids) (28); and previous anticoagulant therapy (5). (Myxœdema was a reason for exclusion, but no patient had this condition.)

Allocation to Test or Control Groups

After selection for the trial on the basis of these criteria, 199 patients were allocated at random, within hospitals, to the diet regimen under test and 194 to the control group.

Description of Patients and Comparison of Groups at Entry to the Trial

Table I compares the two groups in terms of information available at entry to the trial (taken as the first outpatient visit). There was no significant difference in any of the factors examined with the exception of height, the controls being significantly shorter than the men on the test diet ($0.01 < P < 0.02$). With so many factors it is not surprising that one should show a significant difference at this level. Apart from height, which

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TABLE I—COMPARISON BETWEEN PATIENTS ON TEST DIET AND CONTROLS AT ENTRY TO TRIAL

Factor	Test diet %*	Control %*
Angina present before first infarct†	32	26
First infarct graded "severe"†	14	18
Absent peripheral pulses†	11	17
Intermittent claudication†	8	4
Phlebitis or venous thrombosis†	3	5
Arcus senilis, complete ring†	10	7
Aged 50 and over	65	59
Parent died aged 40–64	44	49
Height under 5 ft. 6 in.	23	33
Weight 180 lb. and over	20	19
Serum-cholesterol 325 mg. per 100 ml. and over	15	18
Systolic blood-pressure 160 mg. Hg and over	8	12
Diastolic blood-pressure 100 mg. Hg and over	13	11
Cigarette smokers	81	84
Smokers of 30 or more cigarettes per day	20	20
Social classes I and II†	24	29
Sedentary occupation§	64	62
All patients	199	194

* % of those for whom factor was recorded.
† Diagnosis before randomisation to the two regimens.
‡ Classification of the Registrar General (1960).
§ Classification developed for the Social Medicine Research Unit (Morris and Crawford 1958).

is considered later in terms of relapses, it is clear that randomisation was effective.

Interval Between Infarction and Entry to the Trial

79% of the patients entered the trial within six weeks of the onset of their first infarction, the shortest interval being twelve days and the longest ninety days. There was no significant difference between the two groups.

DIETS

Dietary Regimen

As far as possible, saturated fats were removed from the diet. Patients on the test diet were instructed to take 3 oz. (85 g.) of soya-bean oil daily. This oil was chosen because it is highly unsaturated and, when used previously in a similar diet, had been shown to cause a satisfactory fall in serum-cholesterol (Pilkington et al. 1960). The soya-bean oil from various sources used by the patients was analysed and found to be a fully standardised product.

At least 43 g. of soya-bean oil daily had to be taken unheated, and it was often drunk with fruit juice. The remainder could be used in cooking. In 10 patients who developed intolerance to the oil, such as nausea and diarrhoea, corn oil was substituted. Up to 35 g. of other fat per day was also allowed. 14 g. of this was taken as a moderately unsaturated margarine ('Blue Band'). Foods allowed daily included lean meat (up to 85 g.), any fish, skimmed milk, and clear soups. Foods forbidden included butter, other margarines, cooking-fat, other oils, fat meat, whole milk, cheese, egg yolk, and most biscuits and cakes.

Control Diet

Patients in the control group ate the diet they would ordinarily have taken.

Reducing Diets

174 of the men were placed on reducing diets for a time during the trial because of persistent weight gain or for slight obesity at the start of the trial. Of this number, 163 were put on a reduced carbohydrate diet (73 in the test group and 90 controls). Only 11 were on a strict 1000 C. reducing diet (5 on the test diet and 6 controls). In the 5 on the test diet the oil was reduced from 85 to 57 g. per day.

ASSESSMENT OF RELAPSE

A review committee, which met every three or four months and consisted of five members (K. P. B., W. W. B., J. A. H., J. N. M., C. W.), assessed relapse according to the following criteria:

A. *Definite*.—(a) Development of new Q waves with or without a history of chest pain (Q waves which developed slowly as a result

of the original infarct in patients on anticoagulants were not accepted as evidence of relapse); (b) rest pain for more than one hour with raised serum-enzyme levels or ischaemic T-wave changes indicating deterioration; or (c) death with post-mortem evidence of recent infarction or thrombosis.

B. *Probable*.—(a) Rapidly increasing angina or the sudden onset of steady-state angina both with ischaemic T-wave changes indicating deterioration; (b) rapid onset of heart-failure with ischaemic T-wave changes indicating deterioration; (c) an episode of rest pain with arrhythmia and persisting E.C.G. deterioration; or (d) sudden death with history of recent chest pain or E.C.G. changes (but without necropsy).

C. *Possible*.—(a) Rapidly increasing angina without ancillary evidence, or the sudden onset of steady-state angina without ancillary evidence; (b) rapid onset of heart-failure thought to be due to infarction but without ancillary evidence; (c) history of rest pain for one hour or more without ancillary evidence; (d) a history of spontaneous rest pain of less than one hour with persisting E.C.G. deterioration; or (e) sudden death without a necropsy.

D. *Other ischaemic cardiac death*.—(a) Sudden death with a post-mortem which revealed no new thrombosis or infarction; or (b) death which was considered to be due to ischaemic heart-disease but in which there was no evidence of further cardiac infarction.

The registrars (research assistants) presented to the review committee information about all patients in whom there was any symptom or sign of a relapse. The evidence consisted of clinical notes, all E.C.G.s, X-rays, and reports of laboratory investigations. 174 patients were reviewed by this committee, 79 of them on the test diet and 95 controls. Of these, 17 and 21, respectively, were considered not to have relapsed. The decisions of the review committee were made "blind" in that the treatment group to which the patient belonged was not disclosed.

The committee also reviewed the criteria for entry in cases where there was any doubt. As a result 15 men in the test group and 13 control patients initially admitted to the trial were removed. (They are not included in the 393 mentioned above or in any of the analyses reported and are additional to the 73 men mentioned previously who did not enter the trial at all.) 15 further men (4 on the diet and 11 control) were excluded from the trial after a period of full participation because of the development of severe hypertension (as already defined in the "Criteria for Exclusion") or malignant disease in 14; the other 1 being withdrawn from the trial by his consultant physician. These men have been included in the analysis up to the date of their removal.

MANAGEMENT

Attendance at Outpatient Clinics

The patient (whether on the test diet or a control) and, where practical, his wife were interviewed by the doctor and dietitian at his first outpatient visit about two weeks after leaving hospital, and the necessary instructions were given. Patients were impressed that they must make every effort to keep their follow-up appointments which, in general, were made every two weeks for the first two months, then at three months and every three months thereafter, though it was not always possible to hold to this plan. Any patient who missed an appointment was contacted. To make sure that relapses were notified as soon as possible each patient carried a card indicating that he should notify the trial registrar if he had chest pain lasting for one hour or more, or if he was admitted to hospital. The patient's hospital notes contained a statement that he was attending a research clinic and asking for notification of a relapse. At each visit he was asked about chest pain, smoking-habits, and work record. His blood-pressure was measured and blood was taken for serum-cholesterol estimation at the three-monthly visits. Serial E.C.G.s were obtained in the early part of the trial (at the first outpatient visit, at one month, and three months from the start) to provide a baseline from which subsequent episodes of E.C.G. deterioration could be judged. Annual E.C.G.s and biennial chest X-rays to determine the cardiothoracic ratio were recorded after the initial period in the trial.

Of the 393 men in the trial, 357 (91%; 182 on the test diet and 175 controls) had good attendance records in that they came to every session as requested during the first two years of the trial and subsequently to at least three of the expected four

attendances each year until the end of the trial or death. In addition 28 men attended fairly well, although some had moved away from London. Only 8 men (3 on the test diet and 5 control) were uncooperative from the start, attending sporadically and then failing to resume contact with the trial despite repeated attempts to see them again.

Dietary Control

Patients in the test and control groups were issued with scales and diet sheets, and asked to record the weight of all food consumed on a different day each week for the first seven weeks after admission to the trial, and thereafter on the first day of each month. These records were used as the basis of the dietary supervision and analysis.

An independent dietitian (J. W. M.) interviewed a sample of patients' wives in each group on one occasion (99 in all, 65 on the test diet and 34 controls). She found little discrepancy between statements made to her and those made to the trial dietitians. There was some evidence that the control patients were eating less fried food than they did before their infarcts.

At the end of the trial the dietitians classified the 199 patients on the test diet into three groups comprising 169 men who cooperated in taking the required amount of oil and who cut down their intake of saturated fat, 5 men who lapsed partially from the diet although they still took some soya-bean oil (the proportion of oil in their diet falling below 50% of the total fat), and 25 men who lapsed completely from the diet. Lapses in the last two categories affected 11% of the total possible time in the trial for the men in the test group.

The average amounts of fat and soya bean oil are given for both groups in fig. 1 in terms of oil and of total fat from all sources, and show remarkable constancy throughout the trial time. The average amount of oil recorded (80 g. per day) and the average amount of other fat recorded daily by the men on the test diet (40 g. per day) are very similar to those in the planned diet (85 g. and 35 g. respectively). The ratio of saturated to polyunsaturated fat in the test diet was approximately 1:2 on average and in the control diet was 6:1. In only two instances do oil and fat consumption seem to have changed after a clinical relapse. 46% of calories were derived from fat on average in the test diet.

Dietary Constituents Other than Oil and Fat

The average consumption of carbohydrates and sugar was remarkably steady, and in each case was slightly higher in the diet group than in the controls. Men in the control group averaged 228 g. per day carbohydrate of which 60 g. per day was sucrose, and the men in the test group ate 243 g. per day carbohydrate which included 66 g. per day sucrose. In this context it must be remembered that more controls than men in the test group were on a reducing diet which involved primarily

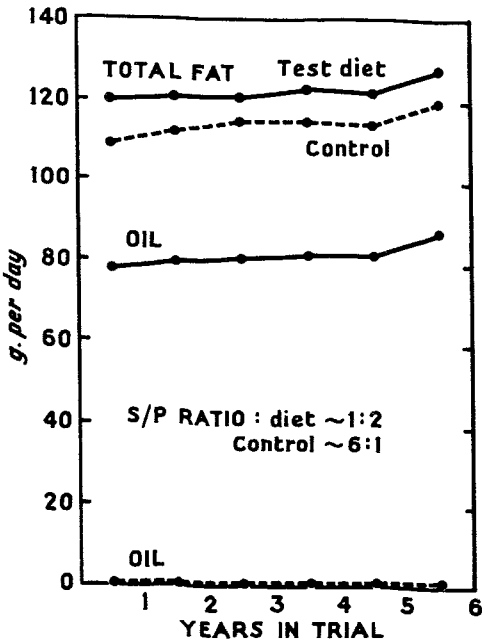


Fig. 1—Dietary fat.

TABLE II—MEAN WEIGHT AND BLOOD-PRESSURE (B.P.)

Time	Weight (lb.)		Systolic B.P. (mm. Hg)		Diastolic B.P. (mm. Hg)		No. of men*	
	Test diet	Control	Test diet	Control	Test diet	Control	Test diet	Control
Entry to trial	164	163	133	133	85	84	193	184
6 mo.	161	162	133	133	83	84	175	173
1 yr.	160	161	132	132	83	84	175	175
2 yr.	161	162	134	133	83	84	168	165
3 yr.	160	162	135	133	85	85	126	126
4 yr.	164	160	135	133	84	87	88	89
5 yr.	164	158	137	137	84	85	48	41

* The figures are the numbers of men for whom weight was determined at each stage. The time given (apart from entry to the trial) is to the nearest three months since patients could not always conform to a rigid programme. The numbers for blood-pressure are very similar. The total numbers in the trial at each stage (using exact time) are shown in the legend to fig. 3.

sucrose and carbohydrate reduction. The average calorie intake per day by the patients on the test diet was 2380 C. and by controls 2274 C.

The consumption of dietary cholesterol was more than twice as great in the controls (588 mg. per day) as in the test group (258 mg. per day). Calcium intake (excluding that from the water supply) was consistently higher in the control diet (930 mg. per day compared with 808 mg. per day for the men on the test diet); this was mainly because most cheeses were forbidden in the test diet. The average protein intake of the men on the test diet was 80 g. per day and of the controls 88 g. per day. A detailed report of the dietary analysis and method of control will appear elsewhere.

Anticoagulant Control

It had originally been intended to compare the effects of long-term and short-term anticoagulation, but unfortunately there was great difficulty in maintaining adequate anticoagulant dosage and uniform prothrombin control in the four hospitals. The anticoagulant part of the trial was therefore abandoned in the 4th year since it was considered that it would not provide a fair comparison of the anticoagulant treatments. The patients (both test and control, except for an early "pilot" group of 27 patients who were all on "short-term" anticoagulation) had been allocated at random, within each hospital, to two anticoagulant regimens—"long term" (or indefinitely prolonged) and "short term" (full anticoagulation for four months from the date of first infarction followed by six weeks tailing-off). This meant that anticoagulation within the diet and control groups was equally distributed, so that a fair assessment could be made of the diet independently of any effect that the anticoagulants might have. It also meant that the anticoagulant treatment in each hospital could be changed without affecting the dietary result provided that, within any hospital, the change was made in everyone concerned. In fact, all patients on long-term anticoagulants were tailed off over six weeks when they had been on anticoagulants for two years or longer, except at Barnet General Hospital where patients on long-term anticoagulants continued on this regimen. This change has no effect on the dietary part of the trial.

Clinical and Laboratory Findings

Blood-pressure

Mean systolic and diastolic blood-pressures are given in table II. There is little difference between the two groups at any stage, and there was no tendency for blood-pressure to rise during the trial.

Weight

The average weight of the patients in each group remained remarkably similar and constant around 160 lb. (73 kg.) in both groups (table II).

Serum-cholesterol

Mean levels during the trial are shown in fig. 2 and

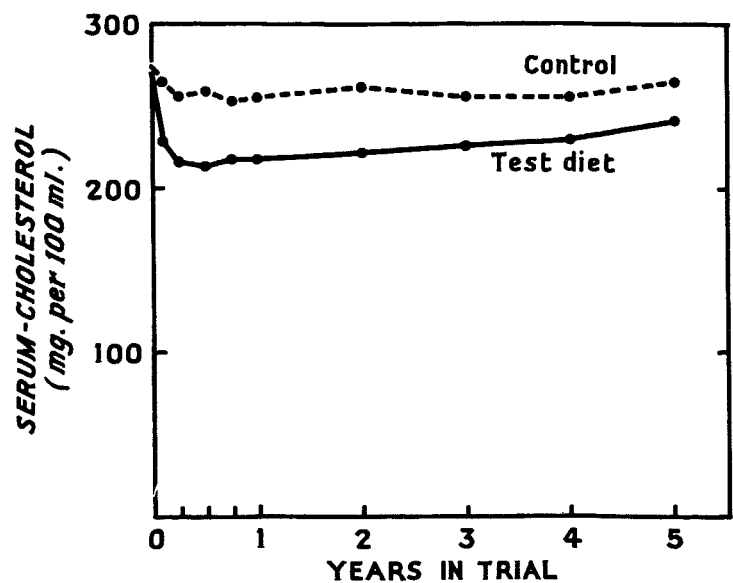


Fig. 2—Serum-cholesterol.

table III. There was an immediate and highly significant fall of 22% (59 mg. per 100 ml.) in the test group to a minimum at six months. There was a steady rise from this low point to the end of the 5th year in the trial when the fall from the starting value was 33 mg. per 100 ml. or 12%. From an almost identical start the level in the control group fell by 6% (14 mg. per 100 ml.) in the first six months and also showed a rise later so that the difference between the groups was maintained.

The cholesterol response of the group on the test diet was affected by the degree of observance of the dietary instructions. Compared with the 169 men who maintained their diet, there was distinctly less fall in the 30 who lapsed from it though their cholesterol level was at all times below that of the control patients. Those 27 men who consistently took less than 60 g. per day of oil had a generally higher serum cholesterol level than those who consistently took more than 60 g. per day.

Serum-cholesterol was estimated by the method of Abell et al. (1952) and later by the automated procedure of Block et al. (1965) as modified by Antonis (1967), giving excellent correlation with the Abell method, the mean difference being +1.4 mg. per 100 ml. These and all chemical estimations were carried out at St. George's Hospital.

Serum-triglycerides (Fasting)

A single estimation was made by the automated fluorometric method of Kessler and Lederer (1965) as modified by Antonis (1967) during 1965 or 1966 on every man then

TABLE III—SERUM-CHOLESTEROL LEVELS AT SUCCESSIVE POINTS IN THE TRIAL

Time	Mean serum-cholesterol (mg. per 100 ml.)		No. of men*	
	Test diet	Control	Test diet	Control
Entry to trial ..	272	273	173	171
6 mo.	213	259	175	174
1 yr.	218	255	177	170
2 yr.	221	262	166	158
3 yr.	226	257	118	127
4 yr.	229	255	89	88
5 yr.	241	265	46	41
6 yr.	239	269	10	12

* Because of lack of information, the number of men shown at any stage is less than the number in the trial. In particular, there are 49 missing values for cholesterol at entry to the trial. For 44 of these men (26 on the test diet, 18 controls) values obtained while the patient was in the ward, and determined by the hospital laboratory, are available. If these are included, the mean values at entry are 268 mg. per 100 ml. for the patients in the test group and 273 mg. per 100 ml. for the controls.

in the trial in two of the hospitals. 110 were on the test diet and 107 in the control group. More of the patients on the test diet (81 men) had serum-triglyceride values below 150 mg. per 100 ml. than the controls (70 men), but the difference was not significant. There was a wide range of individual values (from 37 to 472 mg. per 100 ml.). In a 10% sample, estimations were made on a second occasion. The mean difference between pairs of observations made on the same individual was 33 mg. per 100 ml.

Tissue-fat Biopsy

Fat-biopsy specimens were also taken from 50 patients (28 on the test diet and 22 controls). Results, which will be reported elsewhere, confirm that the tissue fat of the men on the soya-bean oil diet was less saturated than that of the controls.

Relapses and Deaths

Suspected events were classified by the review committee using the criteria listed above under "Assessment of Relapse".

We have analysed these relapses and deaths in three ways: (1) all first relapses; (2) "major" relapses, and (3) all deaths from cardiovascular causes.

TABLE IV—FIRST RELAPSES

Relapse*	Diet group	Control group
A. Definite†		
(a)	10	10
(b)	16	16
(c)	5	5
B. Probable:		
(a)	4	8
(b)	0	0
(c)	0	1
(d)†	4	0
C. Possible:		
(a)	1	4
(b)	1	0
(c)	13	18
(d)	3	4
(e)†	0	2
D. Other ischaemic cardiac death†:		
(a)	4	5
(b)	1	1
All first relapses	62	74

* See text under "Assessment of Relapse".
† Major relapse.

We emphasise first relapses because, at the start of the trial, we had agreed that the main criterion of success or failure of the diet under test would be occurrence of a first relapse, and that after this the clinician responsible for the patient was free to change the treatment. Details are given in table IV. In the event, however, little or no change was in fact made in the patient's diet after a first or later relapse, and we have therefore also considered events in later relapses.

The concept of a "major" relapse was introduced partly for comparison with the Oslo trial (Leren 1966) and also as a useful category of serious events. A major relapse is defined as a "definite relapse" (categories A[a] to A[c]) or a death in any other relapse (categories B[d], C[e], D[a] or D[b]) (table IV).

Results are shown in table V. The number of first relapses is greater, though not significantly so, in the men in the control group (74 compared with 62). The number of major first relapses (fatal or not), however, is virtually the same (39 and 40), and the excess of first relapses in the control group is entirely due to those not classified as major, which include "acquired angina" (35 compared with 22).

TABLE V—RELAPSES AND DEATHS DURING TRIAL

Category	Test diet	Control
No. of men	199	194
First relapses:	62	74
Major	40	39
(a) Fatal	15	14
(b) Non-fatal definite reinfarctions	25	25
Other non-fatal relapses	22	35
Men who had a major relapse at any time during trial:	45	51
Fatal	25	25
Non-fatal definite reinfarctions	20	26
All deaths from cardiovascular disease	27*	25

* 1 death was from subarachnoid hæmorrhage and 1 from cerebral embolism. In addition there were 7 deaths from carcinoma (6 controls and 1 in the test group). The patients concerned were excluded from the trial from the time malignant disease was diagnosed. There were no other deaths during the trial.

More men in the control group had a major relapse at some stage in the trial (51 compared with 45) but again the difference is not significant—i.e., the occurrence of a major relapse as a second or later relapse was commoner in the controls (12 compared with 5).

The total number of deaths ascribed to coronary heart-disease (i.e., fatal major relapses) was the same in each group, and the two deaths from other cardiovascular disease were in the test group, giving a total of 27 cardiovascular deaths in the test group and 25 in the controls.

The number of relapses and deaths in each group takes no account of the stage of the trial at which the relapse occurred nor the numbers at risk at the various stages. "Life-table" analysis, calculating the percentage of men "surviving" (without a relapse or alive) at each stage, based on successive numbers at risk at each stage up to that point, gives results which are in fact very similar to the crude results presented in table v. There is no difference in cardiovascular mortality, little difference in terms of all major relapses, and slightly more difference in terms of all first relapses (fig. 3). In no case is the curve for the test group significantly different from that for the control group.

Relapses and Deaths in Relation to Serum-cholesterol at Entry to the Trial

All men in the trial for whom an initial cholesterol was

recorded (347) are considered in table vi. Apart from the fact that men with cholesterol levels under 225 mg. per 100 ml. seem to have suffered fewer relapses and deaths there is no general tendency for rates to be higher in the men with higher initial cholesterol levels.

Change in Serum-cholesterol in Those Who Relapsed and Those Who Did Not

The cholesterol level for controls who relapsed was consistently slightly higher than for controls who did not, but no such distinction was found in the test group. Similar results are available for deaths and major relapses. Thus, in the test group at least, the amount of cholesterol change was not related to the occurrence of death or a relapse.

Relapses and Deaths in Relation to Dietary Observance in the Test Group

The evidence on this point is conflicting. There is no difference in terms of first relapses, major relapses, or cardiovascular deaths between the 30 patients who were

TABLE VI—RELAPSES AND DEATHS IN RELATION TO SERUM-CHOLESTEROL LEVEL AT ENTRY (MEN IN THE TEST DIET AND CONTROLS COMBINED)

Initial serum-cholesterol (mg. per 100 ml.)	No. of men	First relapses	Major relapses	Cardiovascular deaths
		No.	No.	No.
< 225	61	16 (26%)	7 (12%)	11 (18%)
225-274	128	47 (37%)	22 (17%)	34 (27%)
275-324	98	38 (39%)	17 (18%)	27 (28%)
> 324	60	20 (33%)	5 (9%)	13 (22%)
Total	347	121 (35%)	51 (15%)	85 (25%)

Cholesterol was not recorded for 49 patients at their first visit. Ward values were available for 44 of these men (see footnote to table iii). Inclusion of these makes no material difference to the results.

classified as having lapsed (either partially or completely) from full maintenance of the test diet and the remainder of the test group. Nor is there any relationship between recorded oil consumption and mortality. The first relapse rate, however, showed a consistent trend from a low rate in those who recorded 80 g. of oil or more a day to a high rate amongst the 27 men who recorded less than 60 g. a

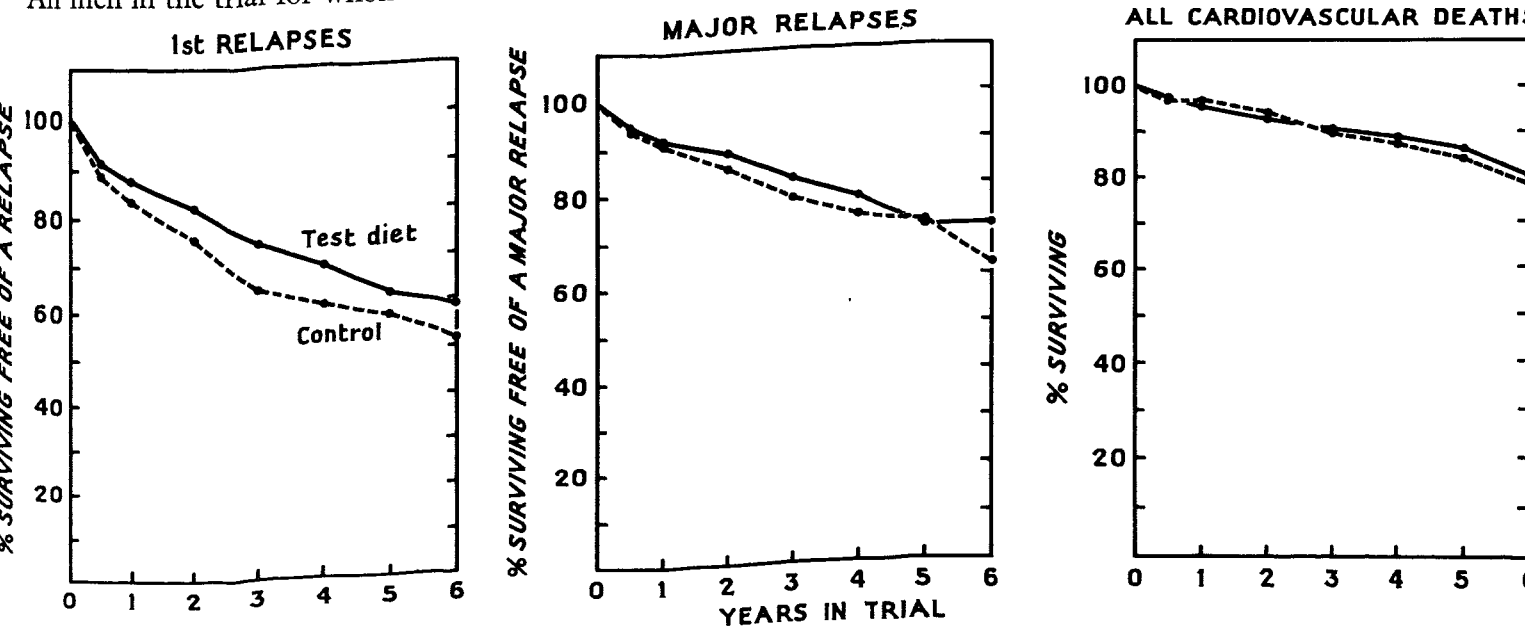


Fig. 3—Percentages free of relapse and surviving, calculated by "life-table" methods.

Numbers surviving in the trial at different stages were as follows:

Stage	Test diet	Control	Stage	Test diet	Control
6 mo.	192	186	4 yr.	91	85
1 yr.	187	185	5 yr.	48	42
2 yr.	181	172	6 yr.	8	9
3 yr.	138	128			

day. This last rate was, paradoxically, higher than in the control group.

Discussion

The design and supervision of clinical trials in men who have survived a myocardial infarct has proved difficult, and the majority of such trials are open to criticism. Oliver (1967) and the review by the standing subcommittee of the National Heart Foundation of Australia (1967) give full details both of drug trials and of dietary trials designed to lower the serum-cholesterol in order to prevent relapse in ischæmic heart-disease, and we will therefore comment on only three.

A research committee of similar composition to ours found no difference after four years in the relapse-rate or death-rate between two groups of patients who entered a trial of a low-fat diet soon after their first infarct (Research Committee 1965). Controls took their normal diet (129 men) and the test group of 123 men a low-fat diet (40 g. daily). In spite of random allocation to treatment, there was a small but significant difference in initial weight between the groups. Both control and treated groups experienced a fall in serum-cholesterol levels but this was greater in the treated group, particularly in the early stages of the trial. The mean difference in serum-cholesterol levels between the two groups was 28 mg. per 100 ml. at six months and 21 mg. per 100 ml. after two years.

Bierenbaum et al. (1967) report a five-year study in two groups of 50 men aged twenty to fifty with "coronary artery disease", who were given one of two different low-fat diets. A matched control group of 100 men with coronary-artery disease was recruited from industry. The controls had a reinfarction-rate 160% higher and a mortality-rate 233% higher than the men on the two diets. This striking result is statistically significant but difficult to assess since there was no randomisation between the diet and control groups at the start of the trial.

Leren (1966) published the results of a five-year trial of men aged under 65 in Oslo with ischæmic heart-disease using a 2300 C. diet containing 75 g. of soya-bean oil. There were 206 men on this diet and 206 randomly chosen controls who ate a normal Norwegian diet. Comparisons between the present trial and the Oslo trial are summarised in table VII. The main differences are those affecting the conditions at the beginning of the trials. For instance, the mean cholesterol levels in London were considerably lower than in Oslo (and tests have shown that this was not due to laboratory technique); the time from primary infarct to the start of the trial averaged twenty months in Oslo and five weeks in London; 14 Oslo men had had a second infarct before they entered the trial whereas none of the London men had had a reinfarction before entry: and of the Oslo men only 61% were aged under sixty at the start of the trial. There was little dietary supervision of the controls in the Oslo trial and in the test group diets taken by only 17 "especially conscientious" patients were fully analysed. The Oslo groups were well randomised apart from the starting blood-pressures which were slightly higher in the diet group. In London height was the only factor to show a significant difference.

The comparison of the results of the two trials in the men aged under sixty is also shown in the lower half of table VII. Major relapses in our trial have been compared with the sum of "sudden deaths" and "myocardial reinfarctions" found in Oslo. We have also attempted to classify the major relapses in the London trial using the Oslo criteria which (for reinfarctions) are somewhat more conservative. Thus, as table VII shows, some London

reinfarctions would not be accepted as such by Leren. (The reverse is also true, but less often.)

The 6 men on the test diet and 12 controls in the Oslo trial who relapsed but did not have a major relapse were classified as "acquired angina". In contrast to the assessment of "myocardial reinfarction" in that trial, which was made in ignorance of the treatment which the patient was receiving, the assessment of acquired angina was not "blind".

The results as a whole show little difference between the trials at the ages at which they can be compared. Indeed the patterns are remarkably similar. In both, there is little difference in total cardiovascular mortality or in sudden deaths from ischæmic heart-disease, more difference in major relapses (as defined), more still in myocardial reinfarctions alone using the Oslo criteria and most in the remaining, non-major relapses. None of the differences between the men in the test diet and the controls in table VII, however, is significant with the exception of all relapses in the Oslo trial (33/47) and here, already noted, the methods of assessment in the two trials are not comparable. The overall relapse and death rates in the two trials are remarkably similar considering the different exposure-times and the fact that the Oslo criteria cannot be exactly applied to the London data.

The possibility must therefore be considered that the results of both trials are consistent in showing no effect on mortality but some difference in the rate of reinfarction and less serious relapses such as acquired angina. Neither trial is large enough to make even an approximate estimate

TABLE VII—COMPARISON OF LONDON AND OSLO TRIALS

Factor	London		Oslo	
No. of men in trial	393		412	
Aged under 60 at entry	390* (99%)		257 (61%)	
Time from first infarct to entry	12 to 90 days (average 36 days)		13 to 24 mo. (average 20 mo.)	
Second infarction before entry	0		14 (3%)	
Initial cholesterol level (mg. per 100 ml.)				
(a) test diet	272		296†	
(b) control	273		296†	
Soya-bean oil in test diet (g./day)	85		75‡	
Saturated/polyunsaturated ratio in test diet	1:1.8		1:2.4‡	
Time from entry to trial to maxi- mum cholesterol reduction	6 mo. (22% reduction)		3 yr. 3 mo. (21% reduction)	
Period of observation	2-6¾ yr.		5 yr.	
Frequency of outpatient visits	3 monthly		6 monthly	
	Test diet	Control	Test diet	Control
Results for men aged under 60:				
No. of men	199	194	134	123
Relapsed at any time during trial	62	74	33§	47§
Major relapse at any time 	45	51	27	35
Myocardial reinfarction (Oslo criteria)	30	38	17	27
Myocardial reinfarction (London but not Oslo criteria)	7	4	N.A.	N.A.
Sudden death (Oslo criteria)	8	9	10	8
Relapsed, but no major relapse at any time¶	17	23	6	12
All deaths from cardiovascular disease	27	25	18	23

* Three men had their 60th birthday between their first infarct and entry to the trial.
† At all ages; at ages under 60 corresponding figures were 299 and 302 (Leren, personal communication).
‡ Based on diets from 17 "especially conscientious" patients.
§ The only significant difference (at the 5% level) between diet and control groups in this part of the table is in those who relapsed at any time in the Oslo trial ($P < 0.027$).
|| In London, as defined in text; in Oslo, myocardial reinfarctions and sudden death.
¶ In London, probable relapse, B(a)-B(c), or possible relapse, C(a)-C(d), see table IV; in Oslo "acquired angina".

of the size of these benefits. The London trial was designed to detect or exclude with sufficient assurance a 50% reduction in all relapses. This it has done: there was no such reduction. But its results are nevertheless consistent with the possibility of a smaller reduction. A trial to detect or exclude, say, a 25% reduction in all relapses would have had to be about four times as large, and this would have been beyond our resources. Even combining the results of the two trials (which is of doubtful validity) can only produce estimates with large margins of error. For myocardial reinfarction alone, which, combining the two trials at ages under sixty, does show a significant difference ($P < 0.05$), a best estimate of 31% reduction is obtained. Using conventional 95% confidence limits for a difference in proportions, limits as wide apart as 53% and 2% can be roughly estimated for the reduction (i.e., it is possible to be reasonably sure only that the reduction in myocardial reinfarction does not lie outside these limits). For all major events together the best estimate is a 20% reduction, but the corresponding limits are 40% reduction and 5% increase in the risk. To examine myocardial reinfarction alone is reasonable since there are theoretical grounds for supposing that this, rather than death due to electrical instability, is the type of relapse which the diet under test might be expected to prevent. Moreover, in the Oslo trial, the same benefit occurs in men aged over sixty and, in that trial, the difference between the test group and the controls in myocardial reinfarction at all ages is statistically significant.

Not shown in table VII are the relationships between initial cholesterol and relapses in the two trials. Leren (1966) found that initial cholesterol was highly related to subsequent relapses, the higher the cholesterol at the start of the trial the higher the relapse rate. In our trial no such picture emerged (table VI), possibly due to the lower initial cholesterol levels in the men, and the amount of cholesterol change was not related to death or to relapse.

The place of our trial in the various published attempts to prevent relapse by lowering serum-cholesterol using dietary means can now be seen. There could be several factors that influenced the result.

In the first place there could have been deficiencies in the experimental design. The two groups were successfully randomised at the start of the trial (table I) except for height, the controls being significantly shorter. It is unlikely that this masked any effect the diet might have had; rather the reverse, since there is evidence (Morris et al. 1966) that short people are more likely to develop coronary heart-disease. Indeed, in this trial the shorter men had a slightly higher total relapse-rate.

The trial itself was not and could not be "blind". The staff and the patients knew the treatment and there may have been room for bias in the registrars' selection of incidents for consideration by the review committee. To test this, an independent assessment of all the serial E.C.G.'s of the 257 patients who did not relapse (table V) was carried out "blind" by a cardiologist, Dr. John Robinson of the London Hospital, not connected with the trial, who used his own criteria of assessment. He was not told initially of the reasons for his analysis, and at no time were details of the patient's history, diet or clinical investigations made available to him. This check revealed (in Dr. Robinson's opinion) 7 fresh Q waves not previously identified in 137 patients in the test group and 4 fresh Q waves in 120 in the control group. On review of the records there was no note of symptoms in any of these incidents. These results

do not suggest any material bias in bringing cases before the review committee.

The excess of control patients in our trial with nothing but non-fatal, non-major relapses (23 against 17) though not statistically significant is also interesting in this connection. One possibility is that the men in the control group themselves may have felt that they were having less active treatment, and might have been psychologically motivated both to complain more and more readily of cardiac symptoms. The other possibility, of course, is that the diet may have had a real effect in these cases of minor ischaemia.

The diet under test proved effective in reducing serum-cholesterol. Using the formula of Keys et al. (1965) it was calculated that the fall in serum-cholesterol should theoretically have been 55 mg. per 100 ml. In fact the maximum observed fall in the diet group (at six months) was 59 mg. per 100 ml.

Dietary cholesterol reduction such as occurred in the men in the diet group, may have contributed towards reducing the serum cholesterol level. What effect other changes had, for instance a lower intake of vitamin A (1000 I.U. lower in the men on the test diet) and of calcium (122 mg. lower in the men on the test diet), is unknown.

In practical terms, although the men in this trial had suffered a recent infarct and were highly motivated to cooperate fully, some found the diet rigorous and not easy to adapt to life in the office or factory. The idea that a diet with an even greater ratio of poly-unsaturated fats to saturated fats would lower the serum-cholesterol further and therefore possibly act as a better secondary preventive of reinfarction is not easy to justify on practical quite apart from theoretical grounds, at any rate as a routine measure.

Indeed, the results of this trial alone lend little support to this suggestion or to the suggestion that a diet of the kind used should be recommended in the treatment of patients who have suffered a myocardial infarction. Taken together with the results of the Oslo trial there is no indication that this type of diet affects mortality. The combined evidence, however, suggests that a proportion of non-fatal reinfarctions might be prevented, though more evidence is required to confirm this effect, and to make a useful estimate of its size.

It must be emphasised, finally, that these results apply to secondary prevention, that is to the treatment of men who have had an attack of coronary heart-disease. The importance of such diets in primary prevention remains to be assessed.

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INFLUENCE OF GASTRIC EMPTYING-RATE AND OF INSULIN RESPONSE ON ORAL GLUCOSE TOLERANCE IN THYROID DISEASE

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Summary The contributions of changes in the rate of gastric emptying and insulin response to the abnormalities of oral glucose tolerance found in thyroid disease have been studied in 12 thyrotoxic and 9 hypothyroid patients. In thyrotoxicosis, rapid gastric emptying was associated with elevated peak blood-glucose and insulin levels, but there was evidence suggesting some impairment of insulin response. However, in the fasting state, despite normal blood-sugar concentrations plasma-insulin levels were elevated. 2 patients were prediabetic, with diabetic glucose-tolerance curves and markedly impaired insulin release when thyrotoxic; although their glucose tolerance reverted to normal with treatment of the thyrotoxicosis, their insulin responses remained sub-normal. In hypothyroidism gastric emptying was prolonged; blood-glucose concentrations were depressed both fasting and after glucose loading, while insulin levels were unchanged. With the exception of hyperinsulinaemia at the end of the glucose-tolerance tests in the hypothyroid patients, all the abnormalities were reversed by treatment.

Introduction

THE impaired oral glucose tolerance described in thyrotoxicosis and the flat curve of hypothyroidism might be, in part, produced by changes in gastric and small-intestinal function, although this has not been previously directly studied. Abnormalities of glucose metabolism which have been demonstrated in thyroid disease, such as changes in insulin release (Woeber et al. 1966), insulin resistance (Elrick et al. 1961, Hales and Hyams 1964), and altered rates of insulin destruction (Elgee and Williams 1955) might also be involved.

In order to clarify the contribution made by changes in gastric function and insulin response to the abnormalities of oral-glucose tolerance, we have studied the rate of gastric emptying, and simultaneously measured the changes in blood-glucose and plasma-insulin after oral glucose administration, in 21 patients with thyrotoxicosis or primary hypothyroidism. Within-patient comparisons were made before and after treatment.

Methods

11 women and 1 man with thyrotoxicosis (mean age 43 years, range 21–68) and 9 women with primary hypothyroidism (mean

age 47 years, range 18–64) were each studied before and again 3 to 4 months after treatment. The patients, who were ambulatory, were on a liberal carbohydrate diet. The initial clinical diagnosis was confirmed by abnormalities of at least two out of three thyroid-function tests (protein-bound iodine, radioactive-iodine uptake, triiodothyronine resin sponge uptake).

Simultaneous oral glucose-tolerance tests and measurements of gastric emptying were performed with the patient supine on the couch of an automatic scintiscanner fitted with two detectors, one above and one below the table. The rate of gastric emptying was measured by a minor modification of the method of Griffith et al. (1966). A drink of 250 ml. containing 50 g. glucose and 200 μ C of ^{51}Cr chromium chloride, a soluble non-absorbable marker, was given to the patient. The left upper quadrant of the abdomen and lower third of the chest were scanned before the dose and at intervals of a few minutes for 90 minutes. A mark was printed on the scan for every ten pulses recorded by the detectors. The outline of the radioactivity, which in practice largely pooled in the fundus of the stomach, was drawn on the first scan taken after administration of the dose, and this outline was drawn on each subsequent scan. The marks within this outline were counted, and this gave an estimate of the radioactivity in the stomach at each time a scan was obtained. A semilogarithmic plot of stomach radioactivity against time gave a linear relationship and allowed the time taken for emptying of half the stomach contents (the gastric emptying half-time) to be estimated. 2 patients with thyrotoxicosis did not have gastric emptying measurements made.

Samples for blood-glucose and plasma-insulin estimations were taken through an indwelling cannula in an arm vein fasting and at 15-minute intervals for 60 minutes and then each 30 minutes for a further 90 minutes after glucose administration. Blood-glucose was estimated by a ferricyanide method modified for use with an 'AutoAnalyzer'. Plasma immunoreactive insulin was measured by a double-antibody technique (Samols and Bilkus 1964).

Results

Gastric Emptying Half-time (G.E.T._{1/2}) (fig. 1)

The mean G.E.T._{1/2} in the hyperthyroid group was 49 minutes (S.E. 2.9), and this significantly increased after treatment ($P < 0.001$) to 63 minutes (S.E. 3.2). In the hypothyroid subjects, the G.E.T._{1/2} decreased with treatment ($P < 0.05$) from 113 minutes (S.E. 15.1) to 68 minutes (S.E. 5.5).

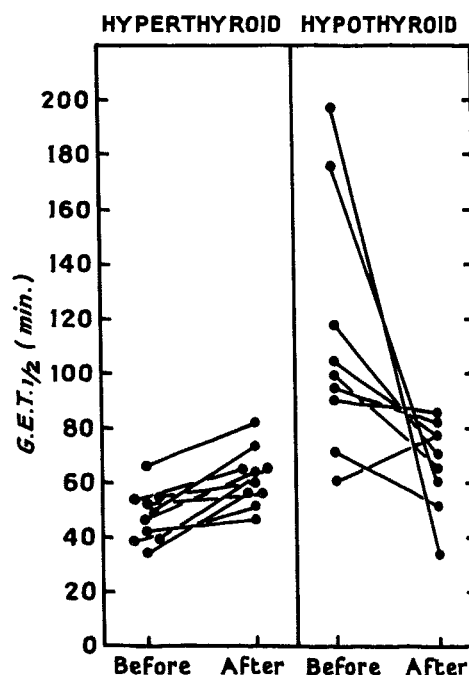


Fig. 1—Change in gastric emptying half times (G.E.T._{1/2}) in hyperthyroid and hypothyroid patients before and after treatment.

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